



ASCO Investor Event

June 3, 2024



Forward-looking statement of Merck & Co., Inc., Rahway, N.J., USA

This presentation of Merck & Co., Inc., Rahway, N.J., USA (the “company”) includes “forward-looking statements” within the meaning of the safe harbor provisions of the U.S. Private Securities Litigation Reform Act of 1995. These statements are based upon the current beliefs and expectations of the company’s management and are subject to significant risks and uncertainties. There can be no guarantees with respect to pipeline candidates that the candidates will receive the necessary regulatory approvals or that they will prove to be commercially successful. If underlying assumptions prove inaccurate or risks or uncertainties materialize, actual results may differ materially from those set forth in the forward-looking statements.

Risks and uncertainties include but are not limited to, general industry conditions and competition; general economic factors, including interest rate and currency exchange rate fluctuations; the impact of pharmaceutical industry regulation and health care legislation in the United States and internationally; global trends toward health care cost containment; technological advances, new products and patents attained by competitors; challenges inherent in new product development, including obtaining regulatory approval; the company’s ability to accurately predict future market conditions; manufacturing difficulties or delays; financial instability of international economies and sovereign risk; dependence on the effectiveness of the company’s patents and other protections for innovative products; and the exposure to litigation, including patent litigation, and/or regulatory actions.

The company undertakes no obligation to publicly update any forward-looking statement, whether as a result of new information, future events or otherwise. Additional factors that could cause results to differ materially from those described in the forward-looking statements can be found in the company’s Annual Report on Form 10-K for the year ended December 31, 2023 and the company’s other filings with the Securities and Exchange Commission (SEC) available at the SEC’s Internet site (www.sec.gov).



Opening Remarks

Dr. Dean Li

President, Merck Research Laboratories



Agenda



Dr. Dean Li

President, Merck
Research Laboratories



Dr. Eliav Barr

SVP, Head of Global
Clinical Development &
Chief Medical Officer



Dr. Marjorie Green

SVP, Head of Oncology
Clinical Development



Chirfi Guindo

Chief Marketing Officer,
Human Health

- **Opening Remarks | Dr. Dean Li**
- **Strategy Overview, Earlier Stage Cancers, IO | Dr. Eliav Barr**
- **Tissue Targeting & Precision Molecular Targeting | Dr. Marjorie Green**
- **Commercial Update | Chirfi Guindo**
- **Closing Remarks | Dr. Dean Li**
- **Q&A | All**

Significant progress since 2021 in providing benefit for patients and delivering upon our strategy

2021 Objective

Leverage KEYTRUDA as a foundation to advance standard-of-care for patients, diversify our pipeline and achieve durable leadership in oncology

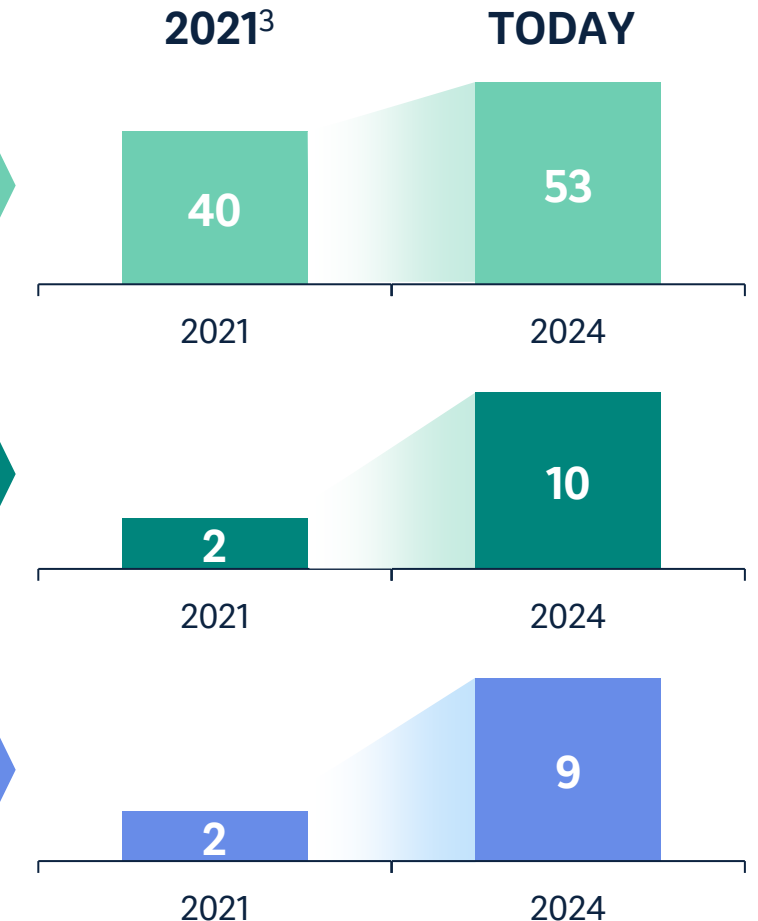
Strategy

Approvals in additional indications¹

Approvals in earlier-stage settings¹

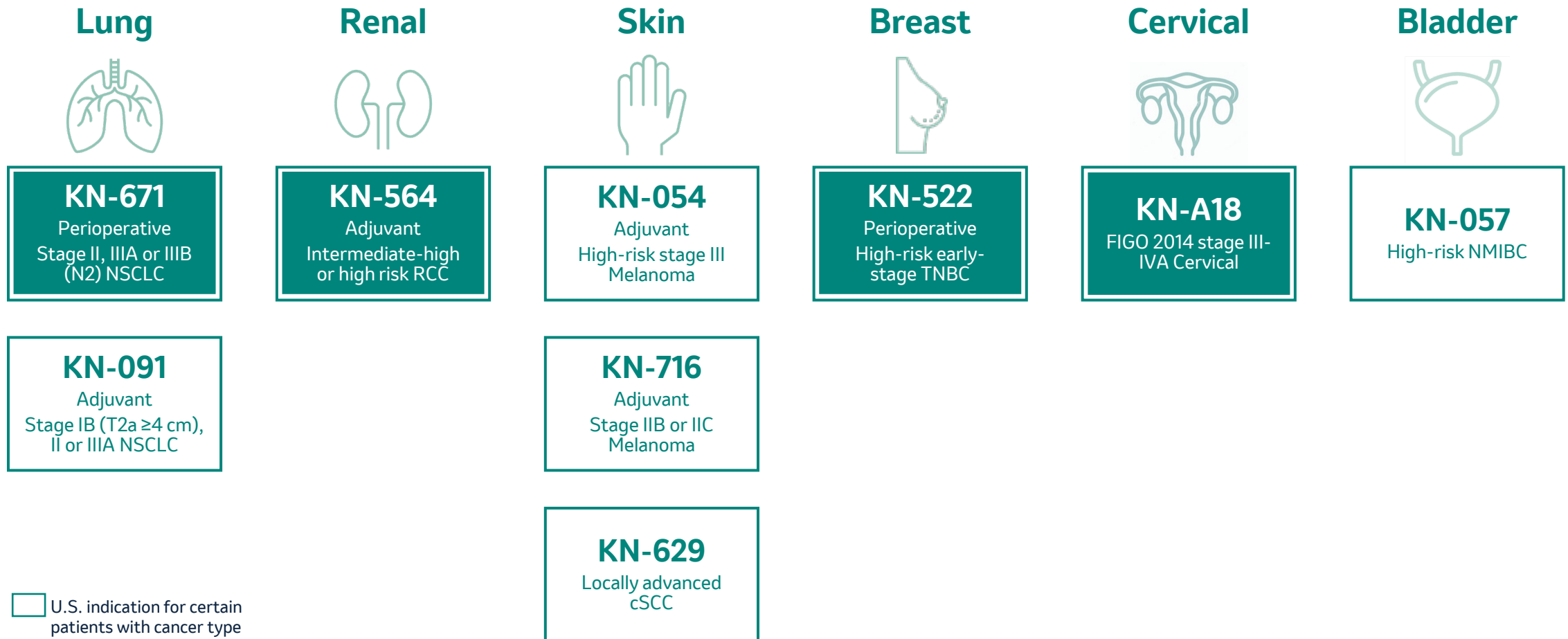
Late-phase candidates studied in combination²

Progress



1. U.S. approvals across oncology portfolio
2. Pipeline candidates being studied in at least one Phase 3 study
3. As of January 1, 2021

In earlier stage disease, KEYTRUDA is the only IO to date to demonstrate significant OS benefit in 4 studies, and to receive 9 approvals



 U.S. indication for certain patients with cancer type

 Achieved OS in cancer type (not all currently labeled)



Strategy Overview, Earlier Stage Cancers, Immuno-oncology

Dr. Eliav Barr

SVP, Head of Global Clinical Development
& Chief Medical Officer



Leveraging solid foundation to build a strong future in Oncology

In 2021 we had...

an immunotherapy-based portfolio ...

... focused on **metastatic and late-stage** disease ...

... building on **pembrolizumab** monotherapy and combination options



Today we have...

a pipeline encompassing **multiple mechanisms of action** ...

... **spanning disease stages** (early stage, metastatic / late-stage) ...

... with **multiple therapeutic options** and candidates for both all-comer and **biomarker-specific patient populations**

Uniquely positioned to advance cancer care by progressing curated pipeline of oncology candidates



Immuno-oncology

Boost anti-tumor immune responses

KEYTRUDA®
(pembrolizumab) Injection 100 mg

hyaluronidase/pembro
(MK-3475A)

V940¹
Individualized
Neoantigen Therapy

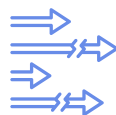
vibostolimab/pembro
(MK-7684A)
anti-TIGIT

favezelimab/pembro
(MK-4280A)
anti-LAG-3

quavonlimab/pembro
(MK-1308A)
anti-CTLA-4

MK-1484
IL-2 R β γ

MK-5890
CD27 agonist



Precision Molecular Targeting

Impact pathways that can drive cancer growth

Lynparza™
olaparib
PARP INHIBITOR

WELIREG™
(belzutifan) 40 mg tablets

nemtabrutinib
(MK-1026)
BTK inhibitor

bomedemstat
(MK-3543)
LSD1 inhibitor

LENVIMA™
(lenvatinib) capsules | 10 mg and 4 mg
RESULTS THAT MATTER

opevesostat⁴
(MK-5684)
CYP11A1 inhibitor

MK-1084⁵
KRAS G12C inhibitor



Tissue Targeting

Increase cancer cell sensitivity with ADCs and immune-engagers

sac-TMT⁶
(MK-2870)
TROP2 ADC

HER3-DXd⁸
(MK-1022)
HER3 ADC

MK-1200⁶
Claudin 18.2 ADC

I-DXd⁸
(MK-2400)
B7H3 ADC

MK-3120⁶
Nectin-4 ADC

R-DXd⁸
(MK-5909)
CDH6 ADC

zilovertamab vedotin
(MK-2140)
ROR1 ADC

MK-6070⁹
DLL3

Undisclosed preclinical ADC targets^{6,7}

Bi- and tri-specific T & NK cell engagers¹⁰

1. Collaboration with Moderna 2. Collaboration with AstraZeneca 3. Collaboration with Eisai 4. Collaboration with Orion
5. Collaboration with Taiho and Astex 6. Collaboration with Kelun Biotech 7. Includes internal pipeline programs 8. Collaboration with Daiichi Sankyo
9. Previously HPN328 10. Collaborations with Dragonfly and Janux and others

More than a dozen novel candidates have advanced to late phase development



Immuno-oncology

Boost anti-tumor immune responses

KEYTRUDA®
(pembrolizumab) Injection 100 mg

hyaluronidase/pembro
(MK-3475A)

vibostolimab/pembro
(MK-7684A)
anti-TIGIT

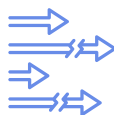
quavonlimab/pembro
(MK-1308A)
anti-CTLA-4

MK-5890
CD27 agonist

V940¹
Individualized
Neoantigen Therapy

favezelimab/pembro
(MK-4280A)
anti-LAG-3

MK-1484
IL-2 R $\beta\gamma$



Precision Molecular Targeting

Impact pathways that can drive cancer growth

Lynparza™
olaparib

WELIREG™
(belzutifan) 40 mg tablets

nemtabrutinib
(MK-1026)
BTK inhibitor

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(MK-3543)
LSD1 inhibitor

LENVIMA™
(lenvatinib) capsules 10 mg and 4 mg
RESULTS THAT MATTER

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Tissue Targeting

Increase cancer cell sensitivity with ADCs and immune-engagers

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(MK-2870)
TROP2 ADC

MK-1200⁶
Claudin 18.2 ADC

MK-3120⁶
Nectin-4 ADC

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Undisclosed preclinical ADC targets^{6,7}

HER3-DXd⁸
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CDH6 ADC

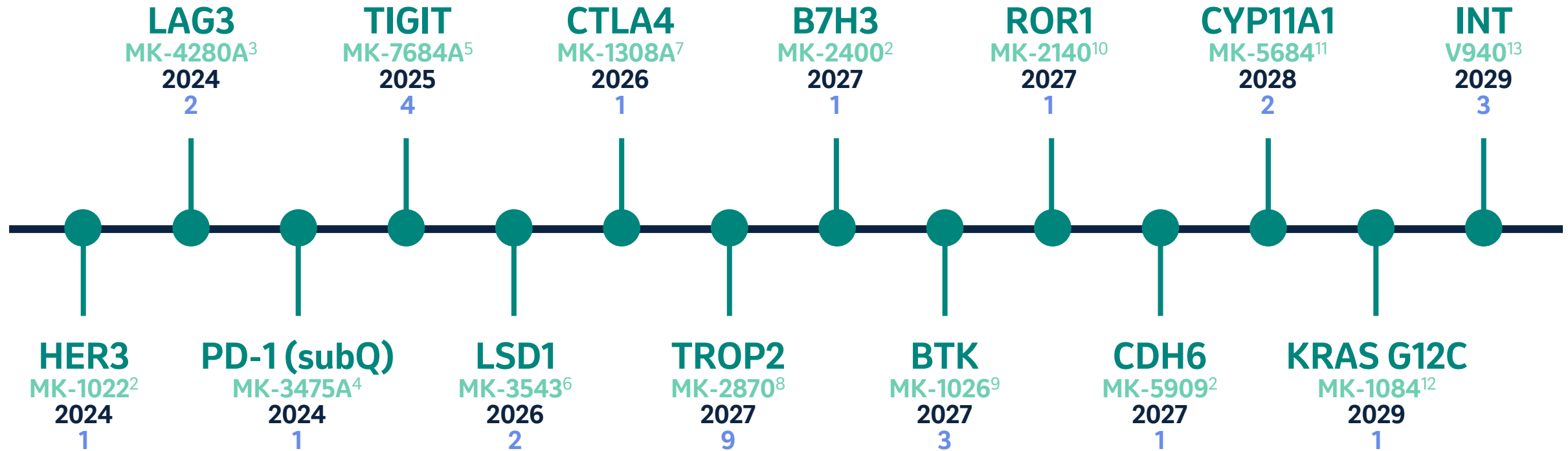
MK-6070⁹
DLL3

Bi- and tri-specific T & NK cell engagers¹⁰

 Late-phase candidates with either an ongoing Phase 3 or Phase 2/3 study

1. Collaboration with Moderna 2. Collaboration with AstraZeneca 3. Collaboration with Eisai 4. Collaboration with Orion
5. Collaboration with Taiho and Astex 6. Collaboration with Kelun Biotech 7. Includes internal pipeline programs 8. Collaboration with Daiichi Sankyo
9. Previously HPN328 10. Collaborations with Dragonfly and Janux and others

>30 ongoing late-phase¹ oncology read-outs expected across pipeline candidates with additional studies being planned



[Target]
[Asset #]
[Earliest estimated PCD]
[# of late-phase studies]

1. Candidates with ongoing Phase 3 or Phase 2/3 study 2. Collaboration with Daiichi Sankyo 3. favezelimab/pembro 4. hyaluronidase/pembro
5. vibostolimab/pembro 6. bomedemstat 7. quavonlimab/pembro 8. sac-TMT, collaboration with Kelun Biotech 9. nemtabrutinib
10. zilovertamab vedotin 11. opevesostat, collaboration with Orion 12. Collaboration with Taiho and Astex 13. Collaboration with Moderna

Extensive ongoing, late-phase clinical development program for earlier stages of cancer¹



Approximately
30,000 patients²
in **>30 active
Phase 3 trials³**
in earlier stages
of disease across
10 tumor types

Esophageal

KN-975

Lung

KN-091 KV-006 INTerpath-002
KN-671 KL-012 TroFuse-019
KN-867 KL-013

Liver

KN-937 LEAP-012

Renal

KN-564 LITESPARK-022

Bladder

KN-057 KN-866 KN-905 KN-B15
KN-123 KN-676 KN-992

Head & Neck

KN-689

Breast

KN-522 KN-242
OLYMPIA TroFuse-012
KN-756

Gastric

KN-585

Skin

KN-054 KN-630
KN-716 INTerpath-001
KN-629

Gynecological

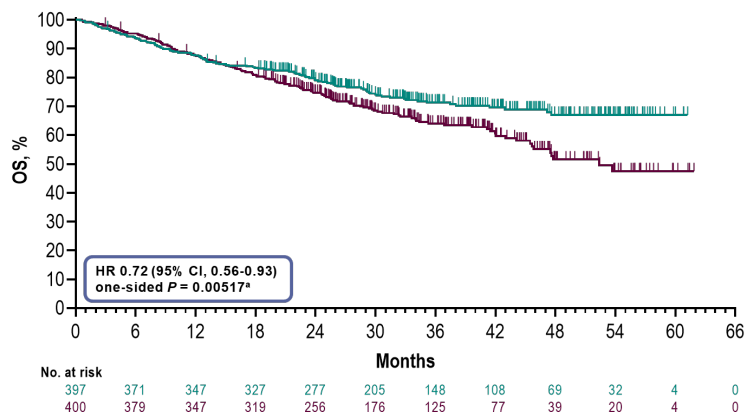
KN-A18 (CERVICAL)

Approved Indication (U.S.)
Announced IA Results

1. Operable or potentially curable 2. Includes targeted number of enrolled patients based upon clinical trial design 3. Reflects ongoing trials including approved studies

KEYTRUDA is the only immuno-oncology therapy to date that has demonstrated overall survival benefit in four earlier-stage cancers

KEYNOTE-671

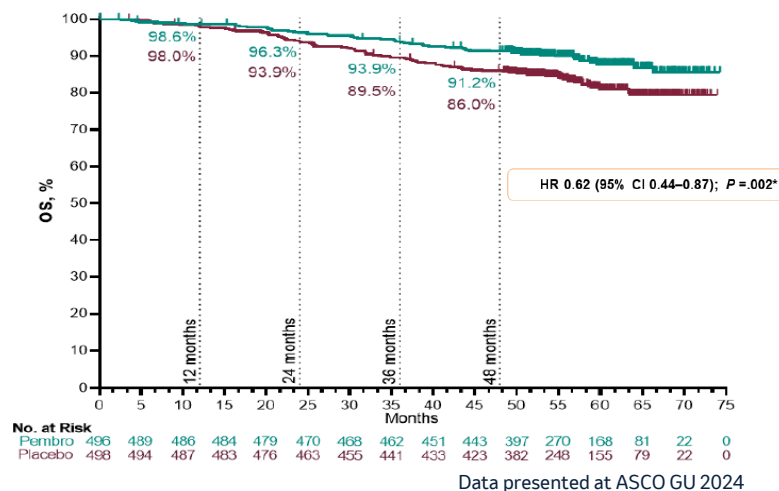


Data presented at ESMO 2023

OS defined as time from randomization to death from any cause. ^a Significance boundary at IA2, $P = 0.00543$. Data reflects results for IA2 - data cutoff date for IA2: July 10, 2023. Median EFS 47.2 months in pembrolizumab arm, 18.3 months in placebo arm

Neoadjuvant KEYTRUDA plus chemotherapy followed by resection and adjuvant KEYTRUDA alone **reduced the risk of death by 28%** in patients with **resectable stage II, IIIA, or IIIB (T3-N42) NSCLC**

KEYNOTE-564



Data presented at ASCO GU 2024

*denotes statistical significance. P-value boundary for OS at IAJ was 0.0072 (1-sided) per Lan-DeMets O'Brien-Fleming spending approximation alpha-spending function. As this key secondary endpoint was formally met, any future OS analyses will be descriptive only. Data cutoff date: September 15, 2023

Adjuvant KEYTRUDA **reduced the risk of death by 38%**, compared with placebo, in patients with resected **clear cell RCC** who were at increased risk of disease recurrence

Women's Cancers

- KEYNOTE-A18:** KEYTRUDA plus chemoradiotherapy **significantly improved overall survival** in patients with newly diagnosed high-risk locally advanced **cervical cancer**
- KEYNOTE-522:** Neoadjuvant KEYTRUDA plus chemotherapy followed by adjuvant KEYTRUDA **significantly improved overall survival** in patients with **high-risk early-stage TNBC**

Building on foundational regimens with novel combination strategies in earlier stage cancers with the goal of improving outcomes for patients

KEYTRUDA-based approvals¹



KN-091

Adjuvant
Stage IB (T2a ≥4 cm), II or IIIA
NSCLC



KN-564

Adjuvant
Intermediate-high
or high risk RCC



KN-054

Adjuvant
High-risk stage III Melanoma

KN-629

Locally advanced cSCC



KN-522

Perioperative
High-risk early-stage TNBC

Ongoing combination studies

V940³

INTerpath-002
Adj NSCLC

MK-2870²

TroFuse-019
Post-neoadj NSCLC

WELIREG

LITESPARK-022
Adj RCC

V940³

INTerpath-004
Adj RCC

V940³

INTerpath-001
Adj Melanoma

V940³

INTerpath-007
Neoadj/adj cSCC

MK-2870²

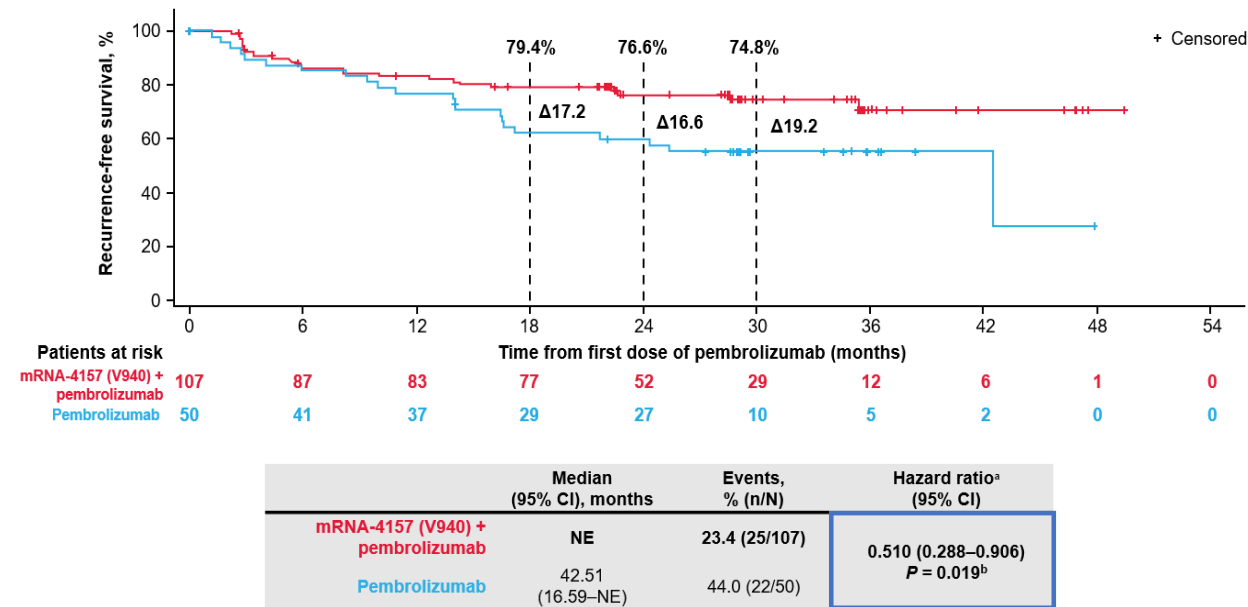
TroFuse-012
Post-neoadj TNBC

Planning
underway for
**additional
combination
studies in earlier
stage disease**

1. U.S. indication for certain patients with cancer type 2. sac-TMT, collaboration with Kelun Biotech 3. Collaboration with Moderna

V940 (mRNA-4157) demonstrated sustained clinical benefit in stage III/IV melanoma

Recurrence-free survival (RFS)



^aThe hazard ratio and 95% CI for mRNA-4157 (V940) + pembrolizumab vs pembrolizumab was estimated using a Cox proportional hazards model with treatment group as a covariate, stratified by disease stage (stages IIIB or IIIC or IIID vs stage IV) used for randomization. The *P* value is based on a 2-sided log-rank test stratified by disease stage (stages IIIB or IIIC or IIID vs stage IV) used for randomization; ^bFormal hypothesis testing of RFS was performed using Nov. 2022 data cut. *p*-value using the Nov. 2023 data cut is nominal and not for a formal hypothesis testing.

- V940 (mRNA-4157) and pembrolizumab demonstrated a **durable clinically significant improvement in RFS & DMFS** vs pembrolizumab in patients with stage III/IV melanoma following complete resection
 - At a median planned follow up of 3 years:
 - 49% reduction in risk of recurrence or death**
 - 62% reduction of distant recurrence**
- Exploratory endpoint of **overall survival** favored the combination¹
- Manageable safety profile** without potentiation of immune-related AEs vs pembrolizumab
- Initiated **Phase 3 studies** for adjuvant treatment in patients with **melanoma** and **NSCLC**, as well as additional earlier-phase studies in other tumor types



Tissue Targeting & Precision Molecular Targeting

Dr. Marjorie Green

SVP, Head of Oncology Clinical Development



Advancing one of the industry's broadest ADC programs

	MK-1022	MK-2870	MK-2400	MK-5909	MK-1200	MK-3120
Target	HER3	TROP2	B7H3	CDH6	Claudin 18.2	Nectin-4
Current Tumor Types ²	EGFRm NSCLC, Breast	NSCLC, Breast	ES-SCLC, Advanced Solid Tumors	Ovarian	GI Tumors	Advanced Solid Tumors
Status	Phase 3 ¹	Phase 3	Phase 3	Phase 2/3	Phase 2	Phase 1
Generic Name	patritumab deruxtecan (HER3-DXd)	sacituzumab tirumotecan (sac-TMT)	ifinatumab deruxtecan (I-DXd)	raludotatug deruxtecan (R-DXd)	Undisclosed	Undisclosed
Partner	Daiichi Sankyo	Kelun Biotech	Daiichi Sankyo	Daiichi Sankyo	Kelun Biotech	Kelun Biotech

**KEYNOTE-A39 /
EV-302³**

Received **FDA approval** for **KEYTRUDA** in combination with **enfortumab vedotin** in 1L **locally advanced or metastatic urothelial cancer**

1. Granted priority review of biologics license application (BLA) in the U.S. based on the Phase 2 HERTHENA-Lung01 data in EGFRm NSCLC that has progressed after EGFR TKI and platinum-based therapies
2. Shows tumor types currently being studied in various phases of development 3. Trial conducted in collaboration with Seagen (now Pfizer) and Astellas

sac-TMT: comprehensive TROP2 development strategy



Registrational studies

Signal finding studies

Biomarker approach

TroFuse-004
EGFRm 3L+

TroFuse-009
EGFRm 2L

First to market opportunities

TroFuse-005
Endometrial

TroFuse-015
Gastric

Novel treatment approaches

TroFuse-019
ES Post-Neoadj
NSCLC
+ pembro

TroFuse-012
Post-Neoadj TNBC
+ pembro

TroFuse-023
NSCLC
(maintenance)

Combination opportunities

TroFuse-007
mNSCLC
PD-L1 TPS ≥ 50%
+ pembro

TroFuse-010
HR+ / HER2-
Metastatic Breast
+ pembro

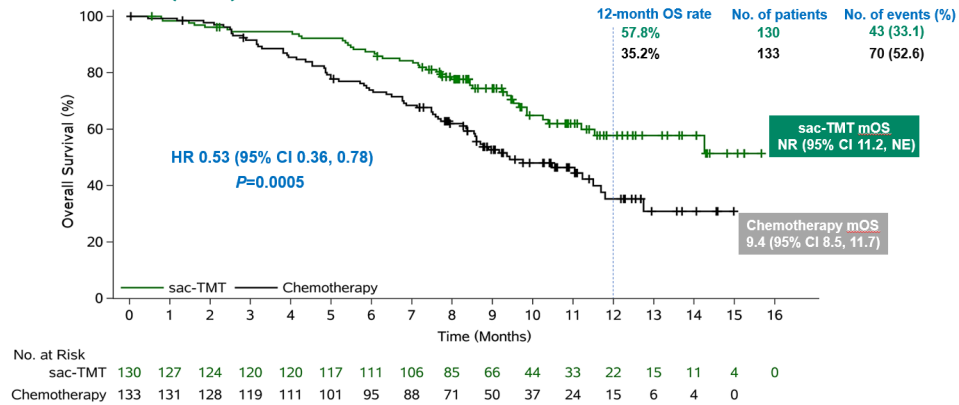
MK-9999-02A
BTC, Pancreatic,
CRC

MK-3475-06B
2L ESCC

Planning for
additional
registrational
and **signal**
finding studies
underway

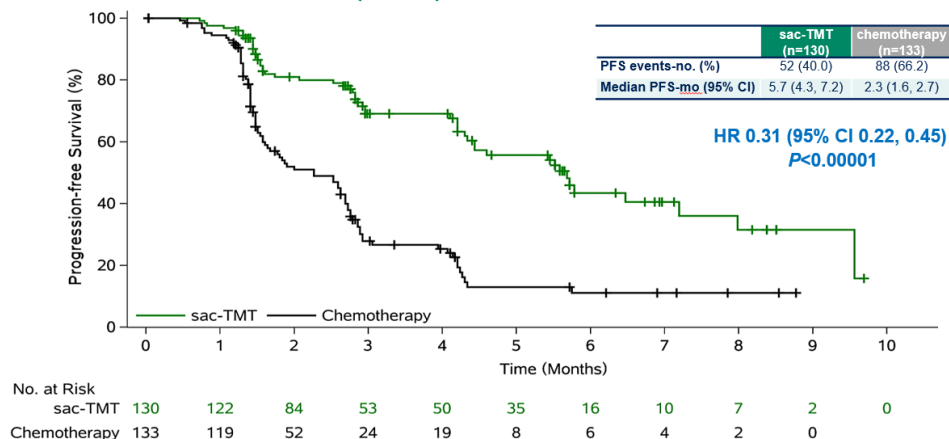
sac-TMT: Phase 3 data primarily in Chinese patients with TNBC demonstrates potential

Overall survival (OS)



Efficacy boundary (corresponding to actual OS events of 113): 0.0042. The study crossed OS efficacy boundary. Data cutoff: Nov 30, 2023, the protocol-specified interim analysis of OS. Median follow-up was 10.4 months. HR, hazard ratio; NE, not estimated; NR, not reached; mOS, median overall survival.

Progression-free survival (PFS)



Efficacy boundary (corresponding to actual PFS events of 140): 0.0104. Data cutoff: Jun 21, 2023, the protocol-specified interim analysis of PFS. Median follow-up was 5.1 months. BICR, blind independent central review; HR, hazard ratio; PFS, progression-free survival.

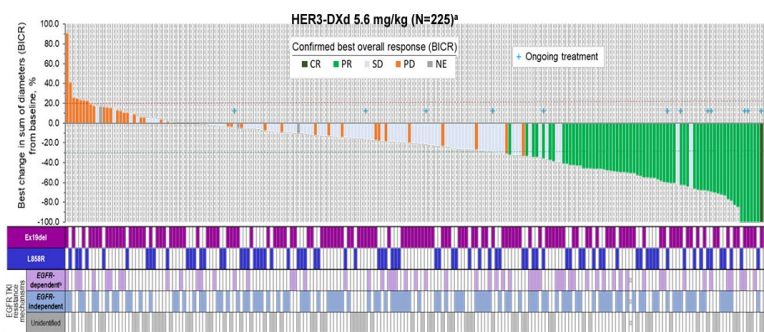
sac-TMT is part of collaboration with Kelun Biotech

- Demonstrated a **statistically significant and clinically meaningful improvement in both OS and PFS** in patients with advanced TNBC vs physician's choice of chemotherapy
- Clinical benefit was observed vs chemotherapy **regardless of TROP2 expression**
 - **Trend for better PFS** associated with patients with **high TROP2 expression** vs low/medium expression
- Showed a **manageable safety profile**

Advancing three potential first-in-class ADCs

MK-1022 (HER3)

Phase 2 data (EGFRm NSCLC)



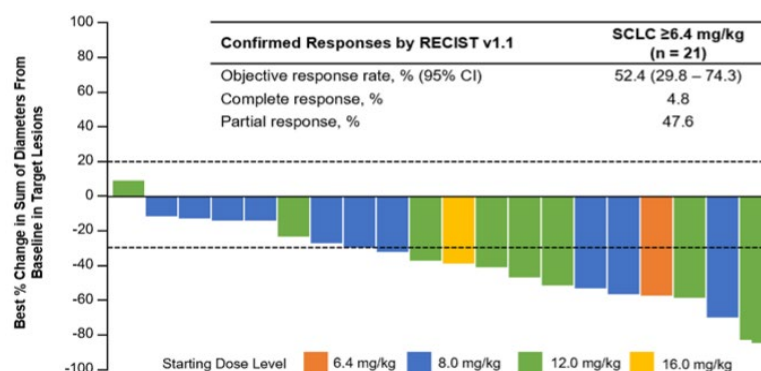
Snapshot data cutoff, 18 May 2023.

Data presented at WCLC 2023

- Received **priority review** in **3L+ EGFRm NSCLC** based on HERTHENA-Lung01 study (PDUFA: June 26th)
- Phase 3 in **2L+ EGFRm NSCLC** as well as **ongoing studies in NSCLC** and **breast cancer**

MK-2400 (B7H3)

Phase 1/2 data (SCLC)



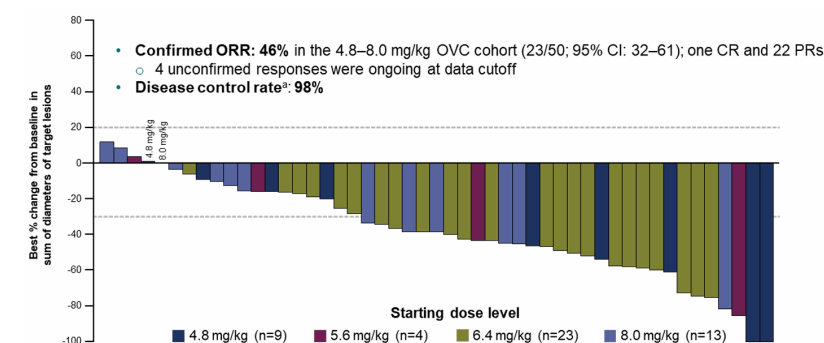
Data cutoff: January 31, 2023

Data presented at WCLC 2023

- Encouraging antitumor activity** observed in FIH study of SCLC, as well as ESCC, mCRPC, sqNSCLC
- Initiated **Phase 3 IDeate-Lung02 study** for relapsed SCLC

MK-5909 (CDH6)

Phase 1 data (ovarian cancer)



Data cutoff data: July 14, 2023

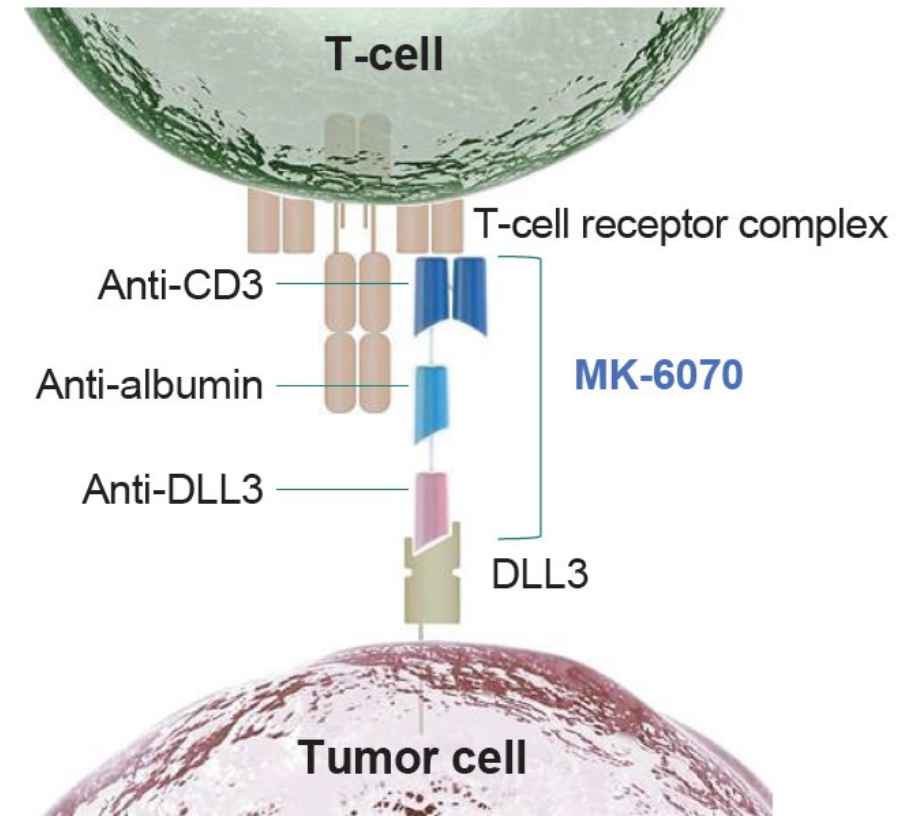
Data presented at ESMO 2023

- Phase 1 data demonstrated **promising efficacy** in patients with heavily pretreated platinum resistant ovarian cancer
- Initiated **Phase 2/3 REJOICE-Ovarian01 study** for platinum resistant ovarian, primary peritoneal or fallopian tube cancer

MK-6070: A DLL3-targeted T cell engager with clinical activity in SCLC and neuroendocrine tumors

- **Acts independently** of MHC expression or co-stimulation
- **Redirects T cells** to kill DLL3-expressing cancer cells
- Small protein (~50kDa) to potentially enable **efficient solid tumor penetration with prolonged half-life**
- Designed to **minimize non-specific T cell activation** and **Fc receptor engagement**, intended to **increase therapeutic window**

Mechanism of action



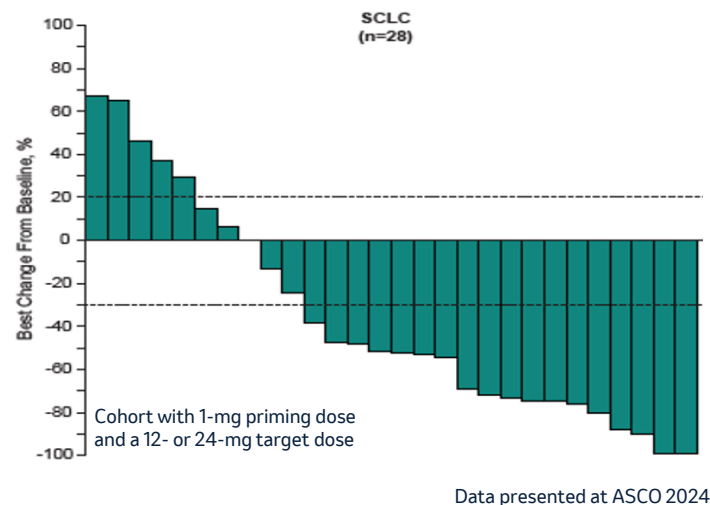
Two highly active agents with potential to improve outcomes for patients with SCLC

- In the U.S., **~34K patients** per year are diagnosed with SCLC¹, representing **~15% of all lung cancers**
- **Median OS** from start of first therapy is **12 months**; **5-year survival** for metastatic SCLC is **~3%**²
- MK-6070 and MK-2400³ are each potentially **highly active** in SCLC, with **manageable safety profiles**

Clinical data

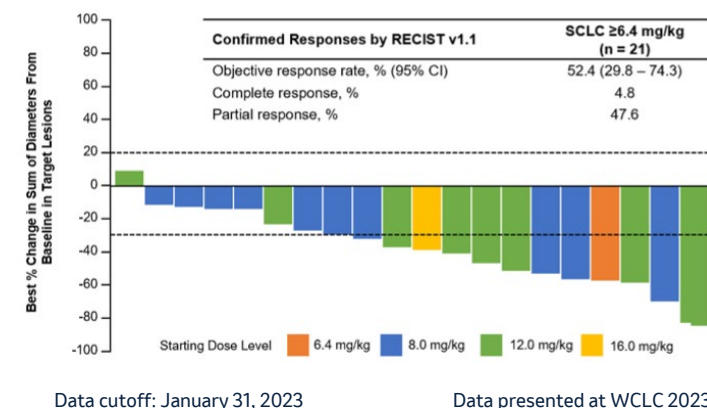
MK-6070

Phase 1 data (SCLC)



MK-2400³

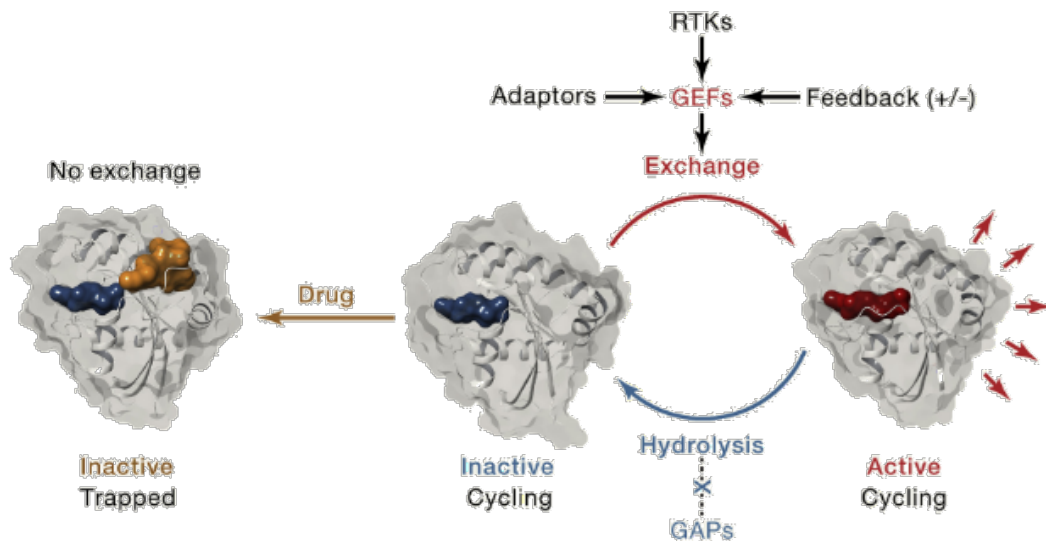
Phase 1/2 data (SCLC)



MK-1084: potential best-in-class combination of KRAS G12C inhibitor with pembrolizumab based on encouraging Phase 1 data

Mechanism of Action

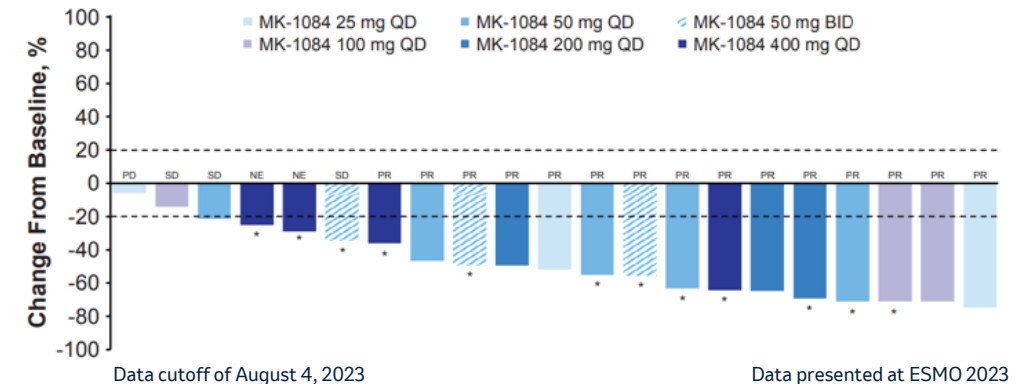
- MK-1084 **selectively targets KRAS G12C mutant and inhibits KRAS G12C-mediated signaling**, which has potential to effectively treat tumors driven by KRAS G12C signaling pathway



Clinical Development

Phase 1 Study - Objective response rate

B. Arm 2 – MK-1084 Plus Pembrolizumab 200 mg Q3W (n = 21)



- MK-1084 in combination with pembrolizumab demonstrated an **ORR of 71%**
 - ORR of 75%** in PD-L1 TPS $\geq$ 50% subgroup
- Advancing to Phase 3** for patients with **KRAS G12C mutant NSCLC with PD-L1 TPS $\geq$ 50%**
- Planning for **additional Phase 3 trials** underway

WELIREG: robust clinical development program evaluating potential across different settings and tumor types



	LITESPARK-004	LITESPARK-005	LITESPARK-011	LITESPARK-012	LITESPARK-022	LITESPARK-015
Phase	2	3	3	3	3	2
Setting	1L	2L+ (post IO & TKI)	2L+ (post IO)	1L	Adjuvant	1L
Treatment	Monotherapy	Monotherapy	+ Lenvatinib	+ Pembro & Lenvatinib	+ Pembro	Monotherapy
Tumor type	Certain VHL disease-associated tumors	aRCC	aRCC	aRCC	RCC	VHL disease-associated PPGL/pNET
Primary endpoint	ORR	PFS OS	PFS OS	PFS OS	DFS	ORR
Status	Approved	Approved (post PD(L)-1 & TKI)	PCD 2024	PCD 2026	PCD 2026	PCD 2027



WELIREG is the **first new treatment** option approved in a novel therapeutic class in advanced RCC since 2015



Commercial Update

Chirfi Guindo

Chief Marketing Officer, Human Health



Driving patient benefits and global growth across a broad portfolio of commercial therapies

KEYTRUDA®
(pembrolizumab) for Injection 50 mg

Foundational cancer treatment in numerous cancers

Lynparza®
olaparib
tablets 150 mg

Market-leading PARPi

WELIREG™
(belzutifan)

First-in-class HIF-2 α inhibitor

LENVIMA®
(lenvatinib) capsules | 10 mg and 4 mg

TKI with multiple approved indications



>2.6 million

patients have been treated with commercially available medicines¹



53 FDA approved indications²

39 approvals in EU
34 approvals in Japan

23 tumor types plus MSI-H / dMMR and TMB-H in U.S.

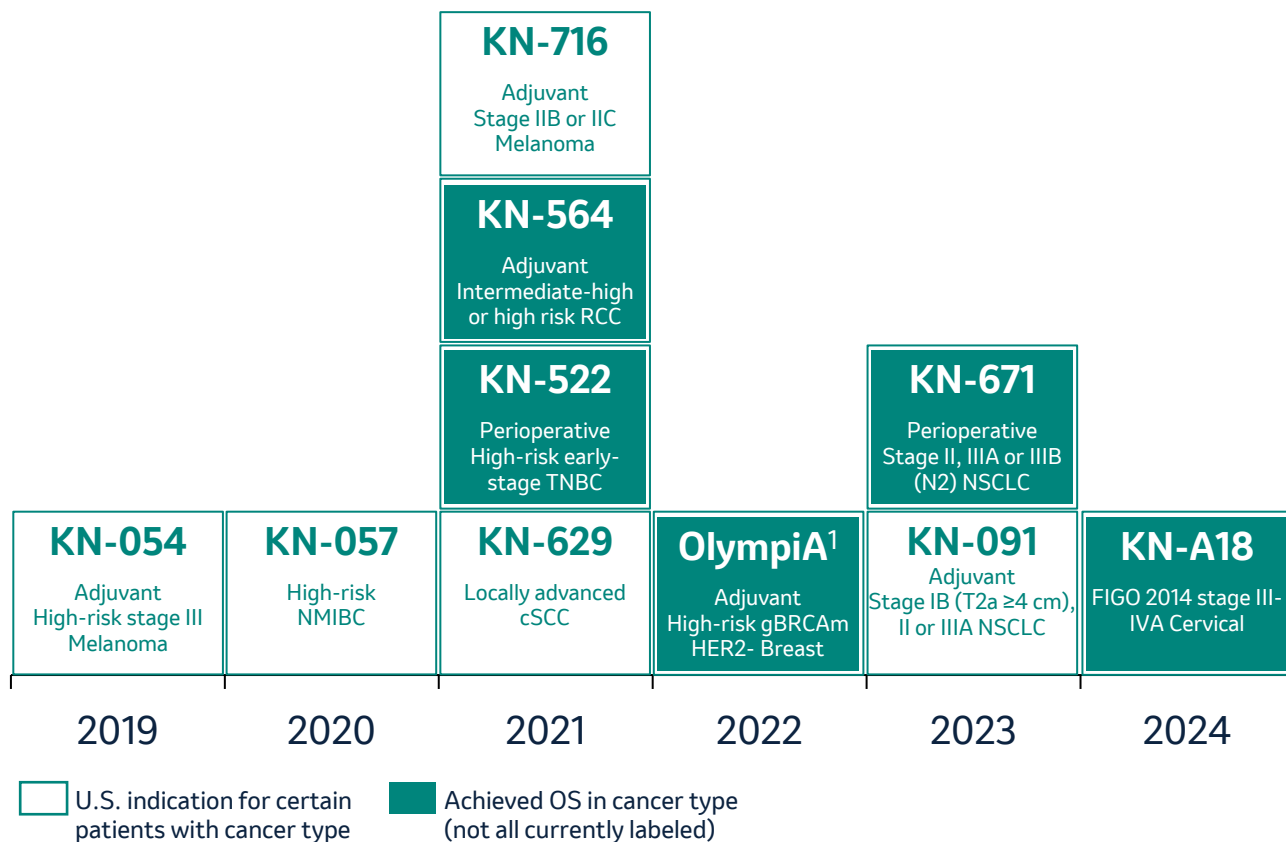
19 tumor types in EU
16 tumor types in Japan

Lynparza and Lenvima are in partnership with AstraZeneca and Eisai, respectively

1. Patients treated with commercially available products as of December 2023 2. BLA and supplemental BLAs

Strong uptake across earlier stage cancer indications based on differentiated clinical profile

10 FDA approvals to date



- **Earlier stage indications** continue to be important **driver of growth**
 - Represented **~20%** of overall KEYTRUDA revenue in **2023**
 - Expected to represent **~25%** of global KEYTRUDA revenue in **2024** and **continue to grow thereafter**
- KEYTRUDA is the only IO to **demonstrate OS benefit** to date in **any earlier stage cancer**
 - Demonstrated **OS benefit** in studies of **TNBC, RCC, NSCLC and cervical cancer**

1. Lynparza in collaboration with AstraZeneca

Establishing broad portfolio in areas of unmet need

NSCLC	Indication	Earlier Stage NSCLC (approved)	mNSCLC (approved)	EGFRm mNSCLC (PDUFA: June 26 th) ¹	EGFRm mNSCLC (earlier lines of therapy)
	Regimen	KEYTRUDA® (pembrolizumab) injection 100 mg -based regimens		HER3-DXd ²	
BLADDER	Indication	Adjuvant MIBC (positive topline results)	NMIBC (approved)	Advanced (approved)	Advanced (additional settings)
	Regimen	KEYTRUDA® (pembrolizumab) injection 100 mg	KEYTRUDA® (pembrolizumab) injection 100 mg -based regimens	KEYTRUDA® (pembrolizumab) injection 100 mg + enfortumab vedotin	KEYTRUDA® (pembrolizumab) injection 100 mg
RCC	Indication	VHL Disease-Associated Tumors (approved)	Adjuvant RCC (approved)	Advanced RCC (approved)	Advanced RCC (additional settings)
	Regimen	WELIREG® (belzutifan) 400 mg tablets	KEYTRUDA® (pembrolizumab) injection 100 mg	KEYTRUDA® (pembrolizumab) injection 100 mg -based regimens	WELIREG® (belzutifan) 400 mg tablets

1. Granted priority review of biologics license application (BLA) in the U.S. based on the Phase 2 HERTHENA-Lung01 data in EGFRm NSCLC that has progressed after EGFR TKI and platinum-based therapies 2. Collaboration with Daiichi Sankyo

Opportunity for patient impact and value creation from a diversified pipeline of innovative programs

Antibody-Drug Conjugates

Individualized Neoantigen Therapy

Small Molecules

**Oncology Pipeline
Opportunity of**

> \$20B

in sales approaching mid-2030s¹

1. Non-risk adjusted annual sales by the mid-2030s; excludes innovation from marketed products, such as fixed combinations, early phase programs or additional business development



Closing Remarks

Dr. Dean Li

President, Merck Research Laboratories



Uniquely positioned to advance cancer care by progressing curated pipeline of oncology candidates



Immuno-oncology

Boost anti-tumor immune responses

KEYTRUDA®
(pembrolizumab) Injection 100 mg

hyaluronidase/pembro
(MK-3475A)

V940¹
Individualized
Neoantigen Therapy

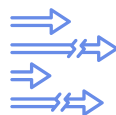
vibostolimab/pembro
(MK-7684A)
anti-TIGIT

favezelimab/pembro
(MK-4280A)
anti-LAG-3

quavonlimab/pembro
(MK-1308A)
anti-CTLA-4

MK-1484
IL-2 R β γ

MK-5890
CD27 agonist



Precision Molecular Targeting

Impact pathways that can drive cancer growth

Lynparza™
olaparib²

WELIREG™
(belzutifan) 40 mg tablets

nemtabrutinib
(MK-1026)
BTK inhibitor

bomedemstat
(MK-3543)
LSD1 inhibitor

LENVIMA™
(lenvatinib) capsules | 10 mg and 4 mg
RESULTS THAT MATTER³

opevesostat⁴
(MK-5684)
CYP11A1 inhibitor

MK-1084⁵
KRAS G12C inhibitor



Tissue Targeting

Increase cancer cell sensitivity with ADCs and immune-engagers

sac-TMT⁶
(MK-2870)
TROP2 ADC

MK-1200⁶
Claudin 18.2 ADC

MK-3120⁶
Nectin-4 ADC

zilovertamab vedotin
(MK-2140)
ROR1 ADC

Undisclosed preclinical ADC targets^{6,7}

HER3-DXd⁸
(MK-1022)
HER3 ADC

I-DXd⁸
(MK-2400)
B7H3 ADC

R-DXd⁸
(MK-5909)
CDH6 ADC

MK-6070⁹
DLL3

Bi- and tri-specific T & NK cell engagers¹⁰

1. Collaboration with Moderna 2. Collaboration with AstraZeneca 3. Collaboration with Eisai 4. Collaboration with Orion
5. Collaboration with Taiho and Astex 6. Collaboration with Kelun Biotech 7. Includes internal pipeline programs 8. Collaboration with Daiichi Sankyo
9. Previously HPN328 10. Collaborations with Dragonfly and Janux and others

Q&A



Dr. Dean Li
President,
Merck Research
Laboratories



Dr. Eliav Barr
SVP, Head of Global
Clinical Development &
Chief Medical Officer



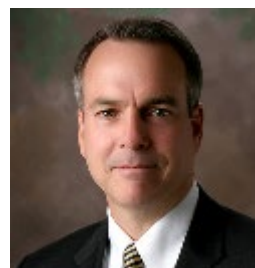
Dr. Marjorie Green
SVP, Head of Oncology
Clinical Development



Chirfi Guindo
Chief Marketing Officer,
Human Health



Joanne Monahan
SVP, U.S. Oncology



Peter Dannenbaum
SVP, Investor Relations



Appendix

Acronyms

ADC = Antibody-drug conjugate

AE = Adverse event

aRCC = Advanced renal cell carcinoma

B7H3 = B7 homolog 3

BCMA = B cell maturation antigen

BTC = Biliary tract cancer

BTK = Bruton tyrosine kinase

CD27 = Cluster of differentiation 27

CDH6 = Cadherin-6

CI = Confidence interval

CRC = Colorectal cancer

cSCC = Cutaneous squamous cell carcinoma

CTLA-4 = Cytotoxic T-lymphocyte associated protein 4

CYP11A1 = Cytochrome P450 Family 11 Subfamily A Member 1

DFS = Disease free survival

DLL3 = Delta-like ligand 3

DMFS = Distant metastasis-free survival

dMMR = Deficient mismatch repair

EFS = Event free survival

EGFR = Epidermal growth factor receptor

EGFRm = Epidermal growth factor receptor mutant

ES = Extensive-stage

ESCC = Esophageal squamous cell carcinoma

GI = Gastrointestinal

HER2 = Human epidermal growth factor receptor 2

HER3 = Human epidermal growth factor receptor 3

HIF-2 α = Hypoxia-inducible factor-2 α

HR = Hazard ratio

HR+ = Hormone receptor-positive

IA = Interim Analysis

IL-2 = Interleukin-2

INT = Individualized neoantigen therapy

IO = Immuno-oncology

KRAS = Kirsten rat sarcoma viral oncogene homolog

LA = Locally Advanced

LAG-3 = Lymphocyte-activation gene 3

LSD1 = Lysine specific demethylase 1

mCRPC = Metastatic castration-resistant prostate cancer

MHC = Major histocompatibility complex

MIBC = Muscle-invasive bladder cancer

mNSCLC = Metastatic non-small cell lung cancer

MSI-H = Microsatellite instability-high

NMIBC = Non-muscle invasive bladder cancer

NSCLC = Non-small cell lung cancer

ORR = Objective response rate

OS = Overall survival

PARPi = Poly-ADP ribose polymerase inhibitor

PCD = Primary completion date

PD-1 = Programmed cell death protein 1

PD-L1 = Programmed cell death ligand 1

PDUFA = Prescription Drug User Fee Act

PFS = Progress free survival

pNET = Pancreatic neuroendocrine tumors

PPGL = Paragangliomas

RCC = Renal cell carcinoma

RFS = Recurrence free survival

ROR1 = Receptor tyrosine kinase-like orphan receptor 1

SCLC = Small cell lung cancer

sqNSCLC = Squamous-cell non-small-cell lung cancer

TIGIT = T cell immunoreceptor with Ig and ITIM domains

TKI = Tyrosine kinase inhibitor

TMB-H = Tumor mutational burden-high

TNBC = Triple negative breast cancer

TPS = Tumor proportion score

TROP2 = Trophoblast cell surface antigen-2

VHL = Von Hippel-Lindau

Dean Li, M.D., Ph.D.

President, Merck Research Laboratories

Dr. Dean Li serves as executive vice president and president of Merck Research Laboratories. He leads the company's worldwide human vaccines and therapeutics research and development organization.

Since joining Merck in 2017, Dean has held leadership roles in the Translational Medicine and Discovery functions and was appointed to President, Merck Research Laboratories in January 2021. Prior to joining Merck, Dean held positions of increasing responsibility in translational medical research at the University of Utah. Most recently he served as the H.A. & Edna Benning Professor of Medicine and Cardiology, chief scientific officer, associate vice president and vice dean at the University of Utah Health System. From 2015 to 2016, he also served as interim CEO of Associated Regional University Pathologists, one of the United States' largest clinical reference laboratories. During his tenure at the University of Utah, he co-founded several biotechnology companies based upon research conducted in his laboratory, including Recursion Pharmaceuticals, Hydra Biosciences and Navigen Pharmaceuticals.

Dean received his Bachelor's degree in Chemistry from the University of Chicago and his graduate and clinical training at Washington University School of Medicine in St. Louis. Dean is a board-certified cardiologist, a member of the American Society for Clinical Investigation and the Association of American Physicians.



Eliav Barr, M.D.

Senior Vice President

Head of Global Clinical Development, Chief Medical Officer

Dr. Eliav Barr is senior vice president and head of Global Clinical Development and Chief Medical Officer at Merck Research Laboratories. He leads all late-stage clinical development for Merck's human health portfolio and pipeline.

Prior to his current role, Eliav led MRL's Global Medical Affairs organization expanding Merck's scientific engagement and implementation efforts in oncology, vaccines and infectious diseases. Since joining Merck in 1995, Eliav has held positions of increasing responsibility including leadership roles in oncology and infectious diseases clinical development. He was also previously Therapeutic Area Head for Infectious Diseases and managed product development teams in Oncology and Infectious Disease.

Eliav is a cardiologist by training. He received his undergraduate degree from Penn State University and his medical degree from Thomas Jefferson University. He completed his Internal Medicine residency and Cardiology Fellowship at Johns Hopkins University, and subsequently pursued post-doctoral training at the University of Michigan. Prior to joining Merck, he held a faculty position at the University of Chicago.



Marjorie Green, M.D.

Senior Vice President

Head of Oncology Clinical Development

Dr. Marjorie Green is senior vice president and head of late-stage oncology at Merck Research Laboratories. She leads all late-stage clinical development programs for oncology.

Marjorie joined Merck from Seagen, where she was senior vice president and head of late-stage oncology, leading clinical development of a diverse portfolio of oncology candidates including multiple antibody drug conjugates. She previously held positions of increasing responsibility at Genentech culminating in her tenure as vice president, Global Product Development, head of breast and gynecologic tumor franchise. Previously, she was assistant professor and medical director of the Nellie B. Connally Breast Center and vice-chair of the Institutional Review Board at the MD Anderson Cancer Center, Houston, Texas. During her tenure at MD Anderson, Marjorie established herself as a nationally recognized clinical expert in the management of breast cancer and the treatment and prevention of associated bone metastases and has authored multiple manuscripts and book chapters on preoperative chemotherapy.

Marjorie received her Bachelor of Arts from the University of Notre Dame and her medical degree from the University of Texas Medical Branch. She conducted an internal medicine residency at University of Virginia School of Medicine and completed fellowships in medical oncology and hematology at the MD Anderson Cancer Center.



Chirfi Guindo

Chief Marketing Officer, Human Health

Chirfi Guindo is chief marketing officer for Merck. He is responsible for leading the development and implementation of the company's long-term strategy for the human health portfolio spanning oncology, vaccines, pharmaceutical and pipeline products.

Prior to this role, Chirfi was executive vice president and head of global product strategy and commercialization at Biogen.

Before joining Biogen in 2017, Chirfi spent more than 25 years with Merck in positions of increasing responsibility in finance, sales, commercial and marketing. During his time with Merck, he led global marketing for Merck's HIV portfolio and also led the company's human health businesses in Canada, the Netherlands and South Africa. Chirfi has been recognized for developing strong talent and forging innovative public-private partnerships that expand access to Merck medicines, while elevating the profile of Merck as a patient-focused company.

Chirfi is a graduate of Ecole Centrale de Paris (France) with a degree in engineering and has a master's of Business Administration from New York University's Stern School of Business.



Joanne Monahan

Senior Vice President, U.S. Oncology

Joanne Monahan is Senior Vice President, U.S. Oncology, Merck. She leads the operations, commercialization and P&L for Merck's U.S. Oncology business, including KEYTRUDA, and oversees partnerships with AstraZeneca and Eisai to jointly market Lynparza and Lenvima.

Beyond oncology, Joanne has had many varied roles during her 30-year career at Merck, including, most recently, spending three years leading the U.S. vaccine business for Merck. Joanne also spent 10 years in primary care, including as Vice President and Global Commercial Lead for the company's cardiovascular and diabetes/NASH portfolios.

In her various roles, Joanne has tackled complex business issues, pioneered the launch of key brands and important indications, led through major market events, and has driven the globalization efforts of important vaccines. For her efforts, Joanne has been recognized with a variety of leadership and marketing awards and received recognition from Merck for her contribution to the business, to customers, and to the growth and development of her people and teams.

Joanne holds a Bachelor of Science in English and Speech Communication from the University of Miami, where she graduated with honors.

